

LAPPEENRANTA, Finland

WMO: 029580

Lat: 61.044N

Lon: 28.151E

Elev: 106

StdP: 100.05

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-20.9	-26.7	0.3	-24.0	-23.1	0.5	-20.7	8.0	-1.9	7.3	-2.4	2.0	10	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	27.3	18.6	25.5	17.6	23.7	16.7	19.8	25.1	18.8	23.5	17.7	22.1	3.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.1	21.9	17.0	12.3	20.9	16.0	11.5	19.7	57.1	25.2	53.7	23.5	50.3	22.2	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.6	6.0	DB	-26.0	29.4	3.8	2.4	-28.7	31.1	-30.9	32.5	-33.0	33.8	-35.7	35.6	(4)
(5)			WB	-26.2	21.1	3.7	1.3	-28.9	22.0	-31.0	22.8	-33.1	23.5	-35.9	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-7.3	-7.1	-2.9	3.9	10.1	14.5	17.9	16.0	10.9	4.6	-0.3	-3.8	(6)	
	DBStd	9.82	6.79	6.61	4.59	3.94	4.36	3.67	3.25	3.03	3.28	3.79	4.38	5.78	(7)	
	HDD10.0	2594	536	479	398	189	53	4	0	1	28	170	308	427	(8)	
	HDD18.3	5005	794	713	657	434	258	125	47	86	222	426	558	685	(9)	
	CDD10.0	692	0	0	0	5	55	140	247	186	57	3	0	0	(10)	
	CDD18.3	59	0	0	0	0	2	11	35	12	0	0	0	0	(11)	
	CDH23.3	458	0	0	0	0	22	86	267	83	1	0	0	0	(12)	
	CDH26.7	84	0	0	0	0	2	12	58	12	0	0	0	0	(13)	
(14) Wind	WSAvg	2.9	2.9	2.8	3.0	2.8	3.0	2.9	2.7	2.6	2.8	2.9	3.1	3.3	(14)	
(15) Precipitation	PrecAvg	645	46	35	38	32	35	53	69	80	65	66	65	57	(15)	
	PrecMax	961	112	107	73	77	111	130	150	157	142	143	150	131	(16)	
	PrecMin	440	9	4	5	5	6	5	0	0	11	15	10	10	(17)	
	PrecStd	111	23	21	18	18	19	28	36	41	34	35	28	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	4.1	9.7	19.1	25.9	28.2	30.3	28.1	21.3	14.1	9.2	7.4	(19)	
		MCWB	2.7	2.4	4.7	11.0	15.4	18.1	19.6	18.9	15.6	11.5	8.4	6.6	(20)	
	2%	DB	2.4	2.3	6.2	16.1	22.9	25.9	28.2	25.7	19.2	12.2	7.8	5.2	(21)	
		MCWB	2.0	1.6	2.8	9.5	14.5	17.2	19.2	17.6	14.7	11.0	7.3	4.7	(22)	
	5%	DB	1.4	1.4	4.2	13.3	20.3	23.8	26.3	23.6	17.6	11.1	6.8	4.0	(23)	
		MCWB	1.1	1.0	1.7	7.8	12.7	15.9	18.5	16.9	13.9	10.1	6.3	3.6	(24)	
	10%	DB	0.5	0.6	2.5	10.7	18.0	21.5	24.4	21.7	16.1	10.1	5.5	2.8	(25)	
		MCWB	0.3	0.3	1.0	6.0	11.3	15.0	17.6	16.3	13.3	9.3	5.2	2.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.6	5.3	12.9	17.5	19.9	21.6	20.6	16.7	12.5	8.5	6.6	(27)	
		MCDB	3.2	3.2	8.9	17.3	23.7	24.9	27.3	25.9	19.2	13.3	8.9	7.2	(28)	
	2%	WB	2.1	1.8	3.4	10.2	15.5	18.5	20.4	18.9	15.6	11.4	7.3	4.8	(29)	
		MCDB	2.3	2.2	5.5	14.8	20.3	23.7	26.0	23.1	18.0	12.2	7.6	5.1	(30)	
	5%	WB	1.2	1.0	2.2	8.2	13.8	17.1	19.5	17.9	14.6	10.4	6.2	3.5	(31)	
		MCDB	1.3	1.4	3.6	12.5	18.7	22.0	24.6	22.0	16.7	11.0	6.6	3.8	(32)	
	10%	WB	0.2	0.3	1.3	6.5	12.2	15.7	18.4	16.9	13.7	9.4	5.1	2.3	(33)	
		MCDB	0.4	0.5	2.2	9.9	17.2	20.3	23.0	20.5	15.7	9.9	5.4	2.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.5	6.8	8.3	9.5	8.7	8.5	8.1	6.9	4.9	3.9	4.3	(35)	
		MCDBR	3.7	4.2	8.5	12.3	13.0	11.8	11.1	10.9	9.0	6.0	4.4	3.8	(36)	
	5% WB	MCWBR	3.7	3.9	5.8	6.9	6.4	5.3	4.4	4.7	5.1	4.6	4.3	3.8	(37)	
		MCDWR	3.6	3.9	6.8	11.3	11.2	10.3	10.1	9.4	8.0	5.3	4.3	3.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.256	0.298	0.323	0.362	0.358	0.355	0.379	0.366	0.341	0.323	0.280	N/A	(40)	
		taud	2.075	2.135	2.213	2.340	2.429	2.468	2.418	2.443	2.500	2.400	2.163	N/A	(41)	
	Edn at Noon		459	665	788	831	866	875	839	821	780	653	426	N/A	(42)	
		Edh at Noon	40	73	95	102	101	99	102	92	73	57	37	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.24	0.82	2.29	3.72	5.09	5.34	5.25	4.03	2.36	0.96	0.26	0.12	(44)	
	RadStd		0.05	0.19	0.30	0.40	0.58	0.35	0.50	0.46	0.26	0.14	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-217	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	-179	-189	N/A	N/A

Nomenclature: See separate page